

MARIA NASEEM

AI Engineer / FDE | AI Solutions Architect | Agentic AI & LLM Systems | Computer Vision | Enterprise Delivery

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PROFESSIONAL SUMMARY

AI Engineer / Forward Deployed Engineer (FDE) and AI Solutions Architect with 10 years of professional experience, including 5+ years architecting and delivering production-grade AI and software systems. Recent work includes real-time video analytics and production counting for an NDA-protected engagement; a four-stage HITL facial-recognition attendance system; a PII tokenization vault with hash-chained audit logging; and a multilingual NLP sentiment platform. Specialise in agentic AI engineering - native MCP servers, multi-agent orchestration (OpenAI / Anthropic / Gemini SDKs), RAG with Qdrant / Pinecone, and Human-in-the-Loop verification gates. Client identities and identifying engagement details are withheld under confidentiality obligations.

TECHNICAL SKILLS

Agentic AI & LLMs: Multi-Agent Orchestration | MCP Servers (built & deployed) | OpenAI Agents SDK | Anthropic Claude SDK | Google Gemini | Grok | LangChain / LangGraph | RAG Pipelines | Embeddings & Similarity Search | Vector DBs (Qdrant, Pinecone, pgvector) | Pydantic structured outputs | HITL controls | Prompt & Context Engineering

Computer Vision: YOLO26n / YOLO11 / YOLOv8 (Ultralytics, custom fine-tuned) | OpenVINO (FP16 inference optimization) | ByteTrack & lap multi-object tracking | Real-time RTSP video pipelines | Object detection, counting & throughput analytics | Camera tamper detection (SSIM, blur, freeze, signal loss) | Facial Recognition (DeepFace, RetinaFace, YuNet + SFace) | OpenCV | PaddleOCR | FAISS vector similarity

NLP & Multilingual AI: NLLB-200 | XLM-R | Aya 23 | PEFT / LoRA fine-tuning | Multilingual & low-resource NLP (Urdu, Roman Urdu, Pashto, Sindhi, Punjabi) | Sentiment / intent / topic / toxicity classification | Kafka streaming pipelines

Backend & APIs: Python | FastAPI | async SQLAlchemy 2.x | Pydantic v2 | Celery + Redis | PostgreSQL (psycopg3, asyncpg, pgvector) | Alembic migrations | WebSockets | JWT/OAuth2 | RBAC | Microservices Architecture | Node.js

Security & Privacy Engineering: PII detection & tokenization (regex, Luhn, spaCy NER) | HMAC-SHA256 | Envelope encryption (Fernet/AES) | Hash-chained audit logs | KMS/HSM key providers & rotation | Dual-gate authorization

MLOps & Observability: Prometheus | Grafana | Alertmanager | structlog | Weights & Biases | DVC | MLflow concepts | LLM evaluation patterns | A/B testing | model versioning | Pytest + pytest-asyncio | GitHub Actions CI/CD

DevOps & Cloud: Docker | Docker Compose | CI/CD Pipelines | Microsoft Azure (Container Apps) | Linux | Nginx | Gunicorn | systemd | certbot | familiarity with Kubernetes patterns

Frontend & Full-Stack: Next.js | React | TypeScript | Tailwind CSS | Streamlit | Playwright (browser automation)

Languages: Python (expert) | SQL | TypeScript / JavaScript | Bash | C++ | PHP

PROFESSIONAL EXPERIENCE

AI Solutions Architect & Full-Stack Developer | Arwen Tech, Islamabad | 2025 - Present

- Architected a real-time video analytics and production-counting system for an NDA-protected engagement. Built RTSP pipelines with YOLO26n (OpenVINO FP16), multi-object tracking, throughput counting, camera health checks, and FastAPI microservices with JWT/RBAC and WebSocket updates. Containerized and deployed on Microsoft Azure Container Apps. Demonstrated on-site; minor updates are in progress before rollout.
- Architected and prepared for deployment a four-stage HITL facial-recognition attendance system for an NDA-protected engagement - enrollment -> candidate retrieval from live RTSP streams -> human verification -> embedding generation and real-time inference. Multi-camera RTSP ingestion with OpenCV SFace embeddings, FAISS index, and confidence-based routing. Validated in an internal on-premises environment.

- Architected and built the Security Vault Service - a bank-grade PII tokenization microservice (detect -> tokenize -> encrypt -> dual-gate access) using regex/Luhn + spaCy NER detection, HMAC-SHA256 deterministic tokens, Fernet/AES envelope encryption with versioned key rotation, and an immutable SHA-256 hash-chained audit log enforced at the database layer. FastAPI + PostgreSQL 16 + Redis/Celery + Prometheus/Grafana observability; 86 automated tests. In internal use at Arwen Tech for active testing and hardening.
- Architected the Multilingual NLP Intelligence Platform - an agentic Kafka pipeline for sentiment, intent, topic, toxicity and sarcasm analysis across Urdu, Roman Urdu, Pashto and other low-resource languages, using NLLB-200 translation and XLM-R/Aya-23 classification, with a human-review loop feeding PEFT/LoRA retraining under shadow-mode deployment. FastAPI + Next.js + PostgreSQL/pgvector + Qdrant; 107 backend tests. Operational internally, running sentiment analysis on live feedback.
- Designed and shipped the Customer Experience Portal - full SLA lifecycle, contract onboarding, and workflow automation. FastAPI + Next.js + JWT + WebSocket live updates, Celery workers, async Postgres, fastapi-limiter rate limiting.
- Delivered additional production microservices: an Agentic Observability Platform (Prometheus -> Alertmanager -> AI Alert Analyzer with a native MCP server, Anthropic Claude SDK + FastAPI); a multi-channel notification orchestrator (email/WhatsApp/FCM, LLM-driven messaging, FastAPI + Celery + Redis); and a Document AI/OCR ingestion service (PaddleOCR + pdfplumber + python-docx). All containerized via Docker Compose with GitHub Actions CI/CD.

Corporate Affairs, Legal Operations & AI Solutions Engineer | Confidential Organization, Islamabad | 2024 - 2025

- AI Solutions Engineering (self-initiated, Forward-Deployed capacity): identified operational inefficiencies in contract and governance workflows and designed an Agentic AI Contract Intelligence Platform - a LangGraph multi-agent system (Python, FastAPI, Claude, RAG on PostgreSQL/pgvector) automating document extraction, semantic search, meeting scheduling, minutes management, and follow-up generation, with confidence scoring and audit logging.
- Corporate Affairs & Legal Operations (primary role): coordinated leadership and stakeholder communication, governance meetings, agendas, minutes, contracts, and regulatory filings. Supported compliance reviews, risk assessments, and cross-functional operations. Identifying client, stakeholder, and jurisdiction details are withheld under confidentiality obligations.
- Delivered a corporate WordPress website and online shop with custom themes, PHP plugins, REST API and third-party integrations, plus contributions to internal booking applications.

Co-Founder & Full-Stack Developer | ZUM Services Providers | 2023 - 2025

- Co-founded ZUM Services Providers and managed full end-to-end product development - building the full-stack platform from scratch while simultaneously running business operations, client acquisition, and service delivery.
- Designed and developed production-ready web applications with RESTful API design, backend business logic in Python/FastAPI, and frontend interfaces - owning the complete software development lifecycle independently.
- Designed and managed PostgreSQL databases - schemas, relationships, and optimized queries using SQLAlchemy ORM.
- Containerized all backend services with Docker and implemented automated CI/CD pipelines via GitHub Actions, ensuring reliable, repeatable production deployments.
- Applied generative AI (ChatGPT) with structured prompt engineering to accelerate marketing and digital content production for the business - an early, practical adoption of LLM tooling to cut content turnaround and cost.
- Delivered solar solution proposals and technical system layouts while leading digital operations - website development, marketing, branding, and digital content creation.

Blockchain Developer | MediaPark | 2020 - 2022

- Designed and delivered scalable native blockchain systems and decentralized applications (DApps) - core architecture (blocks, transactions, consensus validation), ECDSA-based cryptographic signing, and low-level socket-based P2P networking for distributed synchronization and block propagation.
- Developed compliance-oriented smart contract interaction frameworks with controlled access mechanisms, encryption standards, and structured technical documentation to ensure blockchain security and operational integrity.

Earlier Career (2016 - 2020): Internship at IBM Pakistan (2016) - supported PTCL's GPON deployment via PeopleSoft, Siebel, IBM Maximo, and IBM Integration Bus; contributed to GPON training documentation and inventory provisioning. Subsequent operations and IT support roles at Prime Tele Power Solution (2016 - 2017) and Team Work Construction (2018 - 2020).

KEY PROJECTS & OPEN SOURCE

- FTE Sales Lead Engine - Playwright browser automation + AI-driven lead scoring + HITL review layer; reduced manual outreach effort by ~80%.
 - Gesture AI Web App - Real-time hand-gesture detection with Next.js for touchless interaction.
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CERTIFICATIONS & EDUCATION

Master of Information Technology - Quaid-e-Azam University, 2016
Agentic AI Level 2 - Professional | PIAIC / Panaversity, 2026
Agentic AI Level 1 - Developer Fundamentals | PIAIC / Panaversity, 2026
Agent Factory Fundamentals: Building Digital FTEs | PIAIC / Panaversity, 2026 (81%)
Generative AI Applications | Coursera, 2026 (90%)
Build RAG Applications | IBM / Coursera, 2026
AI for Everyone | DeepLearning.AI / Coursera, 2026
Alibaba Cloud Certified Trainer | Bano Qabil, 2026 - In Progress
AI for Cybersecurity Specialization | Johns Hopkins University / Coursera - In Progress